

Mathematics 2, part 4, lecture 2

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- ▶ Linear programming
 - ▶ ~~LP toy problem~~
 - ▶ ~~Problems with LP problems~~
 - ▶ ~~LP duality~~
 - ▶ Dimension analysis
 - ▶ LP interior point methods, why?
 - ▶ GD in interior point methods

Primal and dual problems, recall pitchforks

$$\begin{array}{ll}\max & 3x_1 + 2x_2 \\ & 2x_1 + 2x_2 \leq 480 \\ & 3x_1 + 1x_2 \leq 600 \\ & x_1, x_2 \geq 0\end{array} \quad \text{(P) primal}$$

$$\begin{array}{ll}\min & 480y_1 + 600y_2 \\ & 2y_1 + 3y_2 \geq 3 \\ & 2y_1 + 1y_2 \geq 2 \\ & y_1, y_2 \geq 0\end{array} \quad \text{(D) dual}$$

Sensitivity, meaning of dual variables

- ▶ Is there a meaning to dual variables?
- ▶ ~~Pulling a rabbit out of the hat.~~

Interior point methods

- ▶ How does a combinatorial method work?
 - ▶ initial solutions
 - ▶ stepwise improvements
 - ▶ finishing criterion
- ▶ What is an interior point method?
 - ▶ initial solutions
 - ▶ stepwise improvements
 - ▶ finishing criterion

References

- (A) Kurt Mehlhorn, Sanjeev Saxena, *A Still Simpler Way of Introducing the Interior-Point Method for Linear Programming* (ArXiv, ver. December 8th, 2021)
- (B) Kurt Mehlhorn, Sanjeev Saxena, *A Still Simpler Way of Introducing the Interior-Point Method for Linear Programming* (ArXiv, ver. October 16th, 2018)

Primal and dual problems

$$\min c^T x$$

$$Ax = b \quad (\text{P}), \text{ primal}$$

$$x \geq 0$$

$$\max b^T y$$

$$A^T y + s = c \quad (\text{D}), \text{ dual}$$

$$s \geq 0$$

- ▶ A is $n \times m$ matrix, full row rank, $n < m$
- ▶ x, c, s are m -dimensional
- ▶ y, b are n -dimensional
- ▶ y is **unconstrained**

Weak duality

Claim 1 (Weak duality)

If x is a feasible solution to (P) and (y, s) is a feasible solution to (D), then

(D1) $x^T s = c^T x - b^T y$, and

(D2) $b^T y \leq c^T x$ with equality if and only if $s_i x_i = 0$ for all i .

Reformulation of (P)+(D) as a parameterized problem

$$\begin{aligned} Ax &= b \\ A^T y + s &= c \\ x_i s_i &\approx \mu \text{ for all } i \\ x &> 0, s > 0 \end{aligned} \quad (P_\mu)$$

- ▶ $\mu \geq 0$ is a parameter,
- ▶ note that the cost function is gone,
- ▶ observe the **strict** inequalities.

Goals

- initial solution** For a suitable μ it is **easy** to find a solution (x, y, s) to (P_μ) .
- iterative improvement** Given a solution (x, y, s) to (P_μ) we can find a solution (x', y', s') to $(P_{\mu'})$, where μ' is *much* smaller than μ .
Keep repeating until the parameter μ is sufficiently small.
- final rounding** Given a solution (x, y, s) to (P_μ) where μ is sufficiently small one can extract the solution to both (P) and (D).

Iterative improvement

Let (x, y, s) be a solution to (P_μ) and
let (x', y', s') be a solution to $(P_{\mu'})$ where
 $x' = x + h, y' = y + k, s' = s + f$.

Then

$$\begin{aligned} Ah &= 0 \\ A^T k + f &= 0 \\ h_i s_i + f_i x_i &= \mu' - x_i s_i \text{ for all } i \end{aligned} \quad (S)$$

Properties of iterates

Claim 2

$$(x_i + h_i)(s_i + f_i) = \mu' + h_i f_i \text{ and } (x + h)^T (s + f) = m\mu' + h^T f.$$

Claim 3

$$h^T f = f^T h = \sum_i h_i f_i = 0, \text{ i.e. } h \text{ and } f \text{ are orthogonal to each other.}$$

Claim 4

$$c^T (x + h) - b^T (y + k) = (x + h)^T (s + f) = m \cdot \mu'$$

Invariants in the iteration

In every step we have

(I1) (primal strict feasibility) $Ax = b$ with $x > 0$

(I2) (dual strict feasibility) $A^T y + s = c$ with $s > 0$

$$(I3) \sigma^2 := \sum_i \left(\frac{x_i s_i}{\mu} - 1 \right)^2 \leq \frac{1}{36}.$$

We can take $\mu' = \left(1 - \frac{1}{6\sqrt{m}} \right) \mu$.

Final rounding

Assumption: Let L be a positive number so that every **nonzero** coordinate in an optimal solution of (P) or (D) is at least L .

Lemma

Let (x, y, s) and μ satisfy the invariants (I1), (I2), and (I3). Let x^* be an optimal solution to (P) and let (y^*, s^*) be an optimal solution to (D). Assume that the smallest nonzero value of x^* and s^* is at least L .

- ▶ If $x_i < L/(4m)$ then $x_i^* = 0$ in every optimal solution and
- ▶ if $s_i < L/(4m)$ then $s_i^* = 0$ in every optimal solution.

Theorem (strong duality)

For every i we have $x_i^* = 0$ in every optimal solution or $s_i^* = 0$ in every optimal solution. Thus

$$c^T x^* - b^T y^* = (x^*)^T s^* = 0.$$